NZAMT

NAME:	TEACHER:

1

Elim Christian College

Level 1 Mathematics and Statistics

Practice Examination 2024

91947
Demonstrate mathematical reasoning

Credits: Five

Achievement	Achievement with Merit	Achievement with Excellence
Demonstrate mathematical reasoning.	Demonstrate mathematical reasoning with relational thinking.	Demonstrate mathematical reasoning with extended abstract thinking.

You should attempt ALL the questions in this booklet.

If you need more room for any answer, use the extra space provided at the back of this booklet.

Check that this booklet has pages 2–8 in the correct order and that none of these pages is blank.

Do not write in the grey shaded columns on the right of each page. This area is for markers only.

YOU MUST HAND THIS BOOKLET TO THE SUPERVISOR AT THE END OF THE EXAMINATION.

Assessor's use only

QUESTION ONE

(a) Given that angles in a straight line add to 180°, find the value of x in the diagram below.

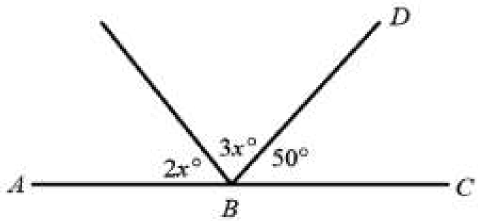
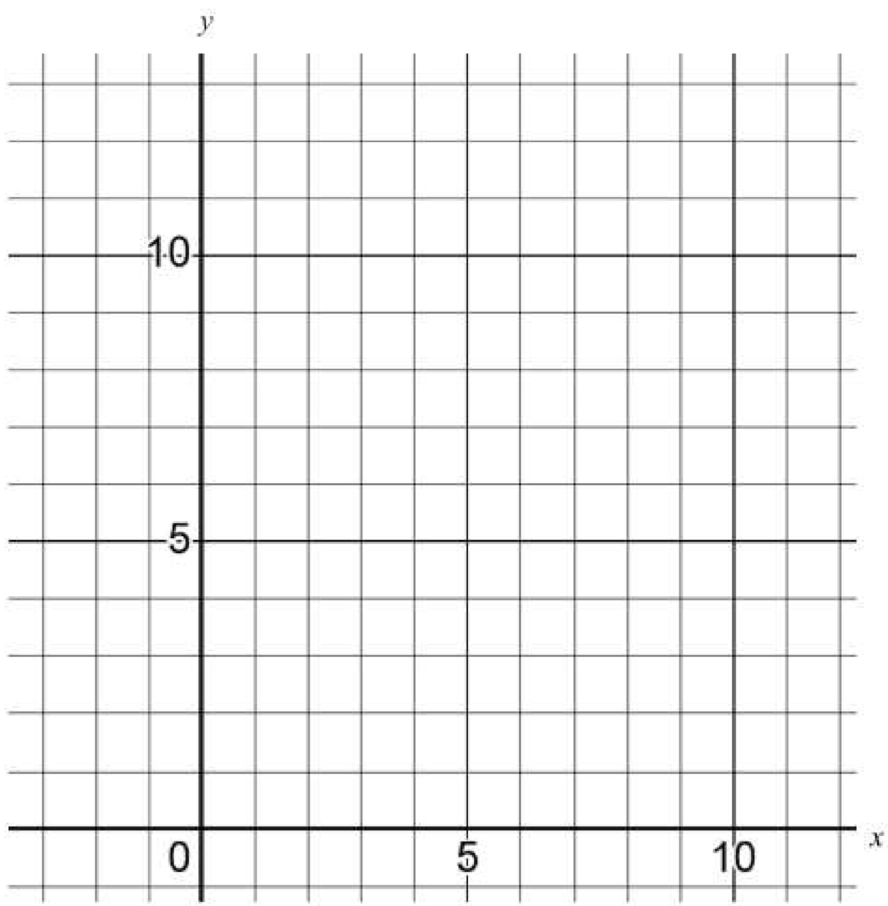


Diagram is NOT to scale

(b) The formula for the height, y metres, of a golf ball above ground level x seconds after it is hit, is given by $y = 7x - x^2$.

i) Find the maximum height reached by the golf ball



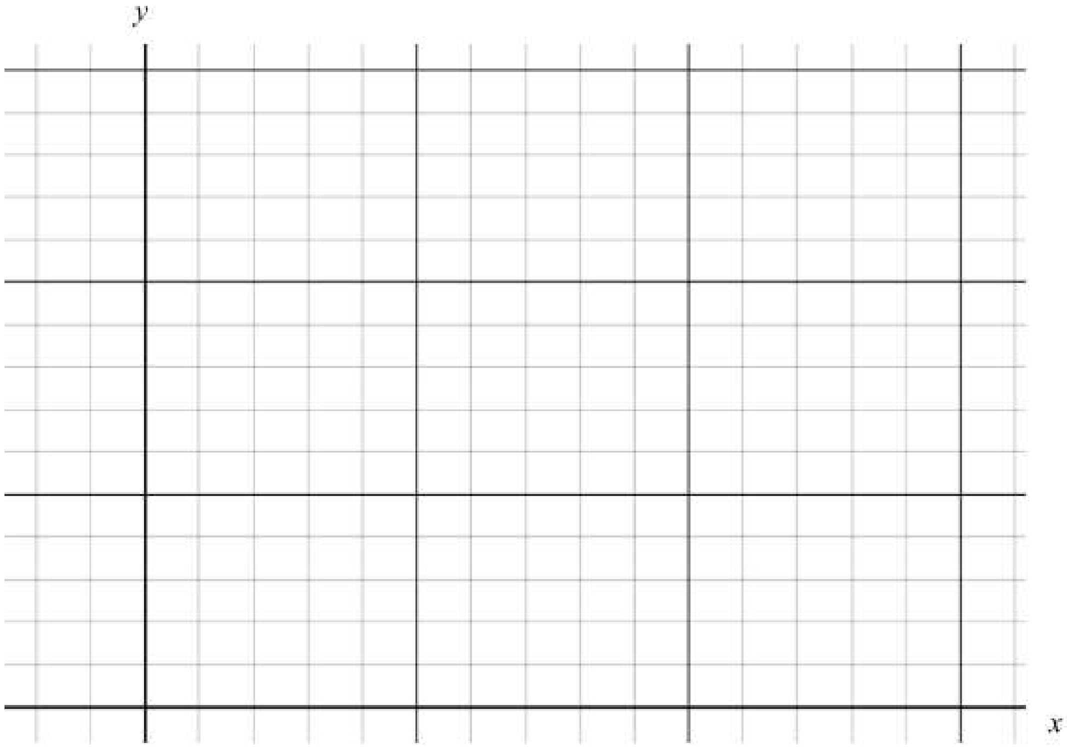
Assessor's use only

ii) Another golf ball landed on the ground 10 metres away from where it was hit and after 4 seconds was 10 metres above ground level.
State the equation that represents the flight of this golf ball.

Assessor's use only

(c) 1000 bacteria are put into a petri dish.
After 1 hour there are 2000 bacteria in the dish.
The bacteria continue to double in quantity every hour.

Using a graph or otherwise, state how long, approximately, it will take the population of bacteria to reach 1 million.



(d) A boat sails due east from the base A of a 30 metre tall lighthouse.
At point B the angle of depression of the boat from the top of the lighthouse is $68°$.
Ten seconds later the boat is at point C and the angle of depression is now $33°$.

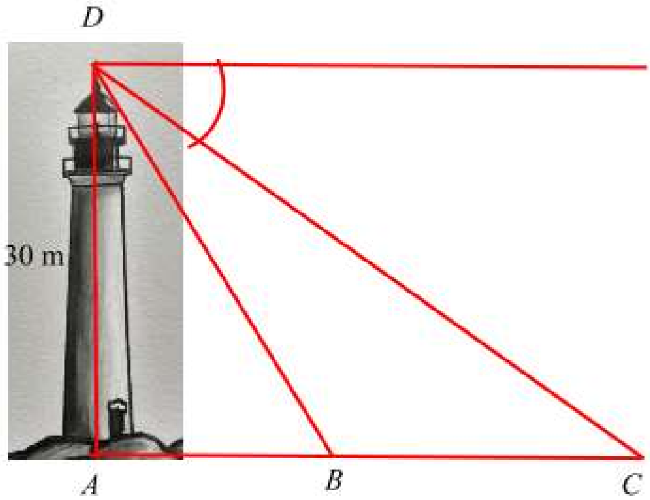


Diagram is NOT to scale

Calculate the average speed the boat is sailing between B and C in kilometres per hour.

Assessor's use only

QUESTION TWO

(a) The diameter of the planet Venus is 1.21×10^4 km.
The diameter of the planet Saturn is 1.21×10^5 km.

What is the difference in length of the diameters of these two planets?
Write your answer in standard form.

(b) $\dfrac{81 \times 27}{3^2}$ can be simplified to 3^n.

What is the value of n?

(c) A steel sphere has a radius of 1.5 cm.

100 of these spheres are melted down and turned into a new bigger cylinder which has a diameter of 10 cm.

Calculate the height of this new bigger cylinder.

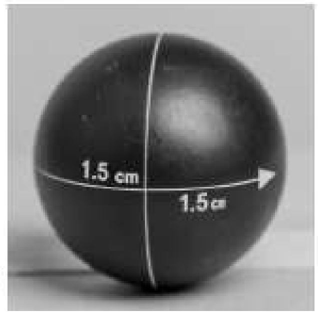
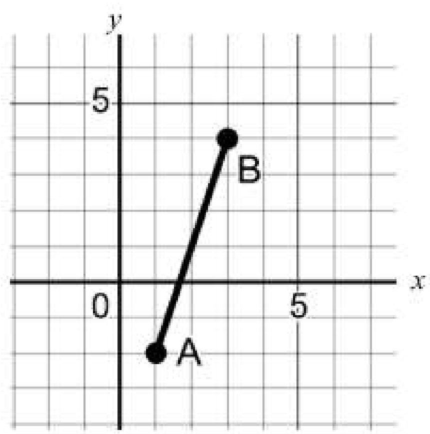


Diagram is NOT to scale

(d) Find the distance between the points $A(1, -2)$ and $B(3, 4)$.



(e) Two circular pulleys with centres 80 cm apart are connected with a tight belt.
The belt wraps $\frac{2}{3}$ of the way around the large pulley which has a radius of 50 cm.
The belt wraps $\frac{1}{3}$ of the way around the small pulley which has a radius of 10 cm.

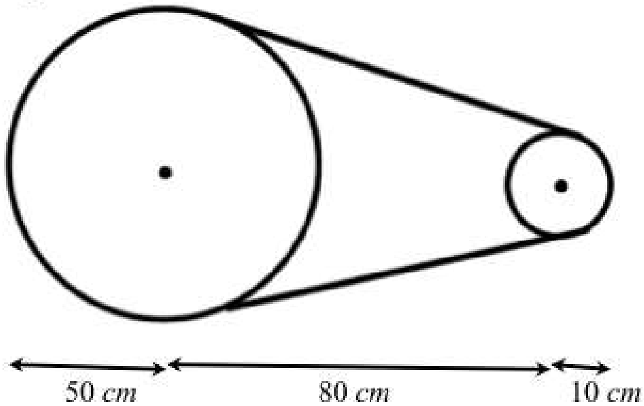


Diagram is NOT to scale

What is the length of the belt?

Assessor's use only

QUESTION THREE

(a) When $m = \frac{2}{5}$ and $n = \frac{5}{4}$, calculate $\frac{1}{2m} - \frac{1}{3n}$.

Write your answer as a fraction.

(b) Solve $\frac{x-1}{3} + \frac{5x+2}{4} = 1$

Assessor's use only

(c) l is the line $2x - 11y = -16$ and k is the line $x + 2y = -8$.

i) Find P, the point of intersection of l and k.

ii) Q (3, 2) is on the line l and R (2, – 5) is on the line k.

Prove that the triangle PQR is isosceles.

Assessor's use only

(d) The diagram below shows an open rectangular box *ABCDEFGH*.
$AB = 18.6\ cm,\ BC = 9\ cm\ and\ CG = 14.5\ cm.$
A straight stick *AGM* rests against *A* and *G* and extends outside of the box to *M*.

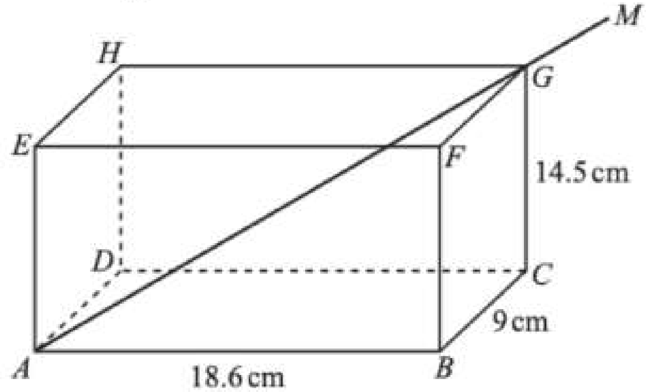
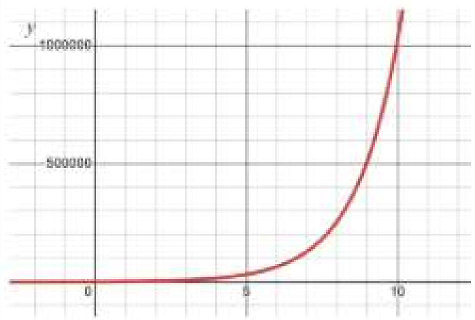


Diagram is *NOT to scale*

The stick *AM* has a length of 30 cm.
Show that the length of $GM = 4.8$ cm.

Assessment Schedule – 2024
Mathematics and Statistics: Demonstrate mathematical reasoning (91947)
Evidence Statement

Q1	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$5x + 50 = 180$ $5x = 130$ $x = 26°$	Correct value of x stated. Accept CAO.		
(b)(i)	$y = 7x - x^2$ $0 = x(7 - x)$ $x = 0, 7$ Turning point: $x = 3.5$ $y = 7(3.5) - (3.5)^2$ $y = 12.25$ m	$x = 0, 7$ stated or CAO	Valid method shown to find height of 12.25 metres	
(b)(ii)	$y = kx(10 - x)$ After 4 seconds was 10 m above ground $10 = k(4)(10 - 4)$ $k = \frac{10}{24} = \frac{5}{12} = 0.416$ So $y = \frac{5}{12}x(10 - x)$ (or equivalent)	Correct set up of equation (recognising x-intercepts) or CAO	Value of k found	Correct function stated
(c)	$1000000 = 1000 \times 2^n$ $2^n = 1000$ So n will be somewhere between 9 and 10 Actual solution: $n = 9.966$ hours (Logarithms are not required to solve this problem)	Exponential function correctly sketched or $1000000 = 1000 \times 2^n$ stated	Any solution beween 9 and 10 hours stated, using numercal reasoning and/or the exponential graph.	

				
(d)	$tan22 = \frac{AB}{30}$ $AB = 12.12\ m$ $tan57 = \frac{AC}{30}$ $AC = 46.20\ m$ $\therefore\ BC = 34.08\ m$ Speed = $\frac{34.08}{10}$ Speed = 3.408 m/s or 12.27 km/h	Length of AB found or Length of AC found or Consistent speed found in m/s	Speed stated in m/s or Consistent speed found in km/h	Speed stated in km/h

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response	1 of N1	1 of u	2 of u	3 of u	1 of r	2 of r	1 of t	2 of t

Note: N1 is an indication of some correct work done in part of a question but not enough to get a u grade – these are judgement calls.

Q2	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$1.21 \times 10^5 - 1.21 \times 10^4 = 108900$ $108900 = 1.09 \times 10^5$	Correct value stated in standard form		
(b)	$\frac{3^4 \times 3^3}{3^2}$ $= \frac{3^7}{3^2}$ $= 3^5$	81 and 27 written as powers of 3 or consistent simplification to find n	$n = 5$ (accept CAO)	
(c)	$\frac{4}{3} \times \pi \times 1.5^3 = 14.137\ cm^3$ $100 \times 14.137 = 1413.72\ cm^3$ $1413.72 = \pi \times 5^2 \times h$ $h = 18\ cm$	Volume of 1 sphere found or Consistent value for h found from incorrect sphere volume	Setting up $1413.72 = \pi \times 5^2 \times h$	Value for h found including units
(d)	$2^2 + 6^2 = 40$ Length of AB = 6.325	Pythagoras consistently used to find an incorrect length	Length correctly stated	
(e)	Length of belt on large pulley: $\frac{2}{3} \times 100 \times \pi = 209.44cm$ Length of belt on small pulley: $\frac{1}{3} \times 20 \times \pi = 20.94\ cm$ Other lengths L_1 & L_2 between pulleys:	Length of belt on large pulley found or	Valid attempt made to find the remaining lengths of belt	Total length of belt correctly found

	$L_1^2 = 80^2 - 40^2$ $L_1 = \sqrt{4800}$ $L_1 = 69.282$ $L_1 + L_2 = 138.56$ Total length = 209.44 + 20.94 + 138.56 = 368.94 cm	Length of belt on small pulley found		

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response	1 of N1	1 of u	2 of u	3 of u	1 of r	2 of r	1 of t	2 of t

Q3	Expected Coverage	Achievement (u)	Merit (r)	Excellence (t)
(a)	$\frac{5}{4} - \frac{4}{15} = \frac{59}{60}$	Correct answer given as a fraction		
(b)	$\frac{4x-4+15x+6}{12} = 1$ $19x + 2 = 12$ $19x = 10$ $x = \frac{10}{19} = 0.526$	Numerator correctly expanded or Consistently solved from 1 minor error	$x = \frac{10}{19} = 0.526$ stated	
(c)(i)	$2x - 11y = -16$ $x + 2y = -8$ $2x - 11y = -16$ $2x + 4y = -16$ $y = 0$ $x + 2(0) = -8$ $x = -8$ Point of int. $P(-8, 0)$	x value found or y value found or CAO	Point of intersection found	
(c)(ii)	$PQ^2 = 11^2 + 2^2$ $PQ = \sqrt{125} = 11.18$ $PR^2 = 10^2 + 5^2$ $PR = \sqrt{125} = 11.18$ Since the lengths of PQ and PR are the same then this triangle must be isosceles.	Length of PQ found or Length of PR found	Length of PQ found and Length of PR found	Statement explaining the reason why the triangle is isosceles
(d)	$AC^2 = 18.6^2 + 9^2$ $AC = \sqrt{426.96}$ $AC = 20.663$	Length of AC found or	Length of AG found	Proof shown as to why $GM = 4.8$ cm

	$AG^2 = 426.96 + 14.5^2$ $AG = \sqrt{637.21}$ $AG = 25.24$ $30 - 25.24 = 4.8\ cm$	AG consistently found from a previous minor error		

N0	N1	N2	A3	A4	M5	M6	E7	E8
No response	1 of N1	1 of u	2 of u	3 of u	1 of r	2 of r	1 of t	2 of t

Cut scores

	Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
Score range	0 – 6	7 – 12	13 – 18	19 – 24